

Grade 7
Unit 3
Lesson 8 Practice

Name Answer Key
Date _____
Period _____

Use the scenario for questions 1, 2, and 3.

Valeria and her friends plan to use a rideshare service for their Prom. The service has a booking fee of \$5.50 and charges \$0.86 for the first $1\frac{1}{2}$ mile and \$0.34 for each additional 0.3 mile after the first $1\frac{1}{2}$ mile.

1. How much will Valeria pay if her trip to Prom is 31.5 miles from her home?

$$31.5 - 1.5 = 30 \text{ additional miles} \quad \frac{30 \text{ mi}}{0.3} = 100 \quad 5.50 + 0.86 + (100 \cdot 0.34)$$

$$5.50 + 0.86 + 34$$

$$\$40.36$$

2. Valeria decides to pick up an additional friend. When they arrive at the friend's house, she is not ready to leave. The rideshare service charges \$1.05 for each minute they have to wait. The total bill was \$46.66. How long did Valeria's friend take to get ready?

$$\begin{array}{r} 46.66 \\ -40.36 \\ \hline 6.30 \end{array} \quad \frac{6.30}{1.05} = 6 \text{ minutes}$$

3. Valeria's friends decide to use the same rideshare service to take everyone home. They drop the first friend off at his home 24.6 miles from Prom. The second friend lives another 6.42 miles and Valeria lives another 4.8 miles away. She decides to tip the driver $\frac{3}{8}$ of the bill. How much did Valeria pay for the service, including tip?

$$\begin{array}{r} 24.6 \\ +6.42 \\ \hline 31.02 \\ +4.8 \\ \hline 35.82 \end{array} \quad \begin{array}{r} 35.82 \\ -1.5 \\ \hline 34.32 \end{array} \quad \frac{34.32}{0.3} = 114.4$$

$$5.50 + 0.86 + (114.4 \cdot 0.34)$$

$$5.50 + 0.86 + 38.896$$

$$45.26 \cdot 1\frac{3}{8} \text{ or } 45.26 \cdot 1.375 = \$62.23 \text{ total with tip}$$

4. Diana purchased $2\frac{5}{6}$ pounds of strawberries, $1\frac{2}{3}$ pounds of blueberries, 1.8 pounds of raspberries and 3 oranges to make homemade fruit salad.

Produce	Price (per pound)
Strawberries	\$2.59 per pound
Blueberries	\$2.65 per pound
Orange	\$0.95 each
Raspberries	\$3.09 per pound

Based on the listed prices, how much would Diana spend?

$$\text{Str: } 2\frac{5}{6} \cdot 2.59 = 7.34 \quad \text{BL: } 1\frac{2}{3} \cdot 2.65 = 4.42 \quad \text{R: } 1.8 \cdot 3.09 = 5.56 \quad \text{O: } 3 \cdot 0.95 = 2.85$$

$$7.34 + 4.42 + 5.56 + 2.85 = \$20.17 \text{ total}$$

Use the scenario for questions 5, 6 and 7.

Isabella offers singing lessons for an initial fee of \$10.50 for the first lesson. Each additional lesson she charges \$6.75 for $\frac{1}{2}$ hour and \$4.86 for each additional 0.33 hour after the first $\frac{1}{2}$ hour.

5. Isabella has a new client this week. The first lesson was held on Monday, then an additional lesson on Wednesday. She received payment totaling \$31.83. How many hours was the lesson on Wednesday?

$$\begin{array}{r} 31.83 \\ -10.50 \\ \hline 21.33 \end{array} \quad \begin{array}{r} 21.33 \\ -6.75 (\frac{1}{2} \text{ hour}) \\ \hline 14.58 \end{array} \quad \frac{14.58}{4.86} = 3 \quad \frac{1}{2} + (3 \cdot 0.33) = 1.49 \text{ or } 1.5 \text{ hours on Wednesday}$$

6. Isabella has a second new client this same week. The first lesson was held on Tuesday, then a second lesson on Thursday for 0.83 hours. On Thursday, how much would she receive for the total singing lessons of this client?

$$\begin{array}{r} 0.83 \\ -0.5 \\ \hline 0.33 \end{array} \quad \begin{array}{l} \text{Half hour} + \text{one } 0.33 \text{ hr} + 1^{\text{st}} \text{ lesson} \\ 6.75 + 4.86 + 10.50 \\ \hline \$22.11 \end{array}$$

7. If Isabella has a third new client this week and only provides the initial singing lesson, how much money in total has she made for the week for all three clients?

$$31.83 + 22.11 + 10.50 = \$64.44$$

8. A cashew store keeps track of the weekly changes in their inventory. Cashews are delivered to the shop in $20\frac{5}{6}$ -pound (lb) bags, 18 bags at a time twice a week. The inventory at the end of week shows a total of $177\frac{3}{4}$ lbs. If the same amount of cashews were sold each day, how many pounds of cashews did they sell in one day?

$$18 \cdot 2 = 36 \text{ total bags} \quad \begin{array}{r} 750 \\ -177.75 \\ \hline 572.25 \text{ lbs. sold} \end{array} \quad \frac{572.25}{7} = 81.75 \text{ lbs per day}$$

$$36 \cdot 20\frac{5}{6} \text{ lbs} = 750 \text{ total lbs.}$$

9. Devon is building birdhouses for the neighborhood birds. To make one birdhouse, he needs 4 pieces of wood $\frac{3}{4}$ feet for the walls, 1 piece of wood 0.25 feet for the floor and 2 pieces of wood $\frac{11}{24}$ feet for the roof. A single board is 8.5 feet long and costs \$10.28. If Devon has \$35, how many birdhouses can he make?

$$4(0.75) + 0.25 + 2\left(\frac{11}{24}\right) = 3 + 0.25 + \frac{11}{12} = 4\frac{1}{6} \text{ ft./birdhouse} \quad \frac{35}{10.28} = 3 \text{ boards she can buy}$$

$$8\frac{1}{2} \div 4\frac{1}{6} = 2\frac{1}{25} = \text{birdhouses per board} \quad 3 \cdot 2 = 6 \text{ birdhouses}$$

10. Bailey is making bows to sell to support her cheerleading team. She purchased a roll of ribbon that is 53.17 ft in length and each bow uses $3\frac{1}{8}$ feet to make. If she makes 10 bows, how many more bows can be made with the ribbon left in the roll?

$$10 \cdot 3\frac{1}{8} = 31\frac{1}{4} \text{ ft used} \quad 21.92 \div 3\frac{1}{8} = 7$$

$$53.17 - 31.25 = 21.92 \text{ ft. left} \quad 7 \text{ more bows.}$$